TIPE 2025 — CYCLE ET BOUCLE

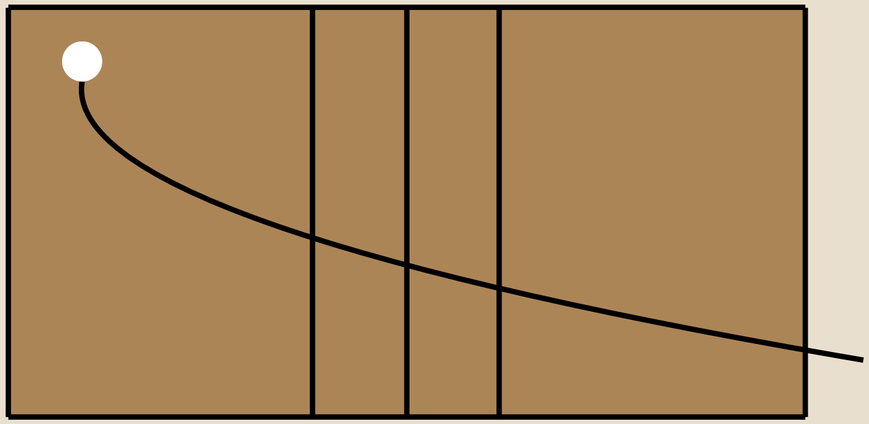
PROBLÉMATIQUE : QUEL IMPACT A L'EFFET MAGNUS SUR UN SERVICE DE VOLLEY ?

BIZID ADAM

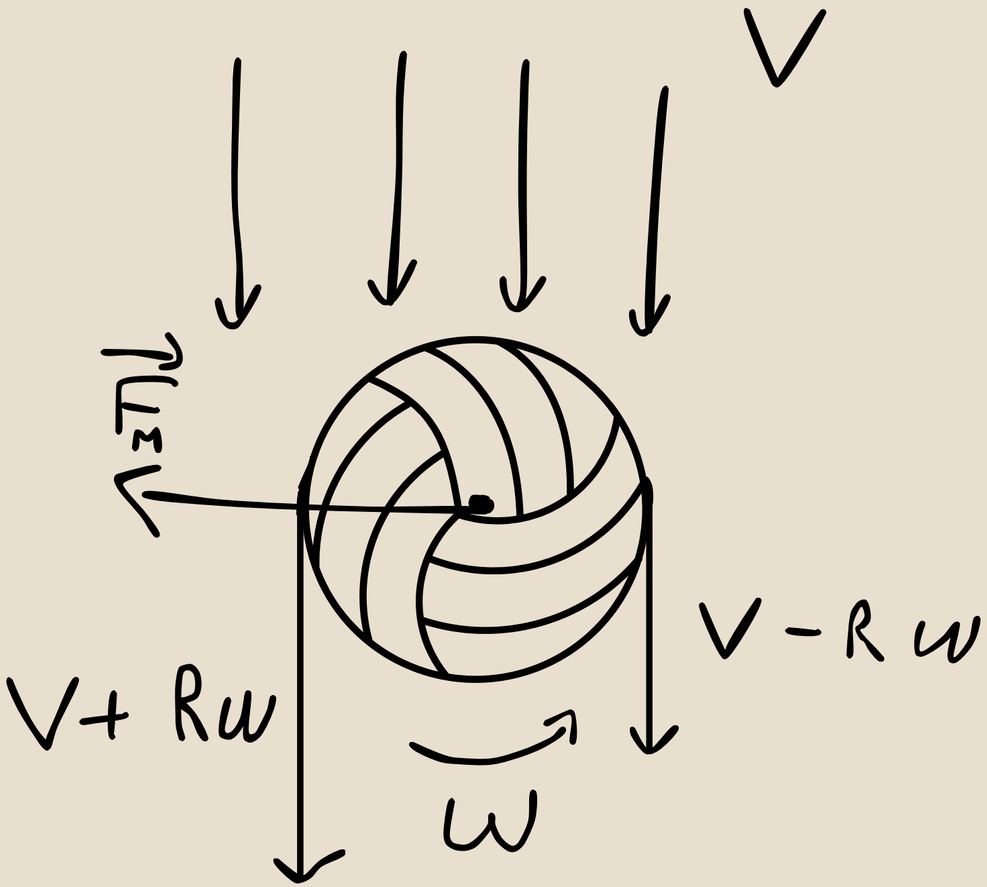
SOMMAIRE

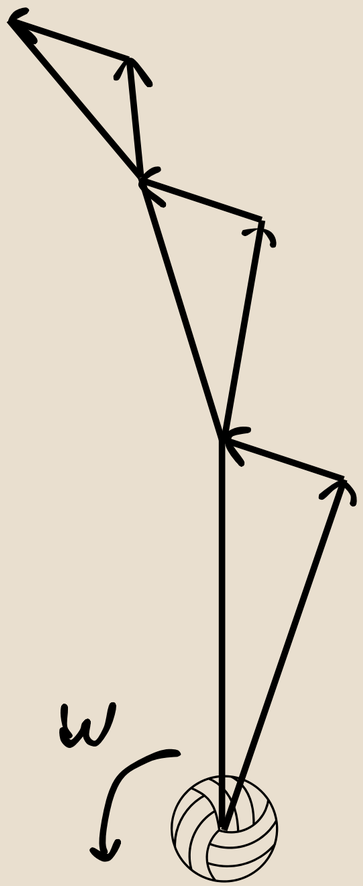
- ÉTUDE EXPÉRIMENTALE D'UN SERVICE DE VOLLEY
- SIMULATION THÉORIQUE AVEC PYTHON
- COMPARAISON ET ANALYSE DES ÉCARTS

DISPOSITIF EXPÉRIMENTAL

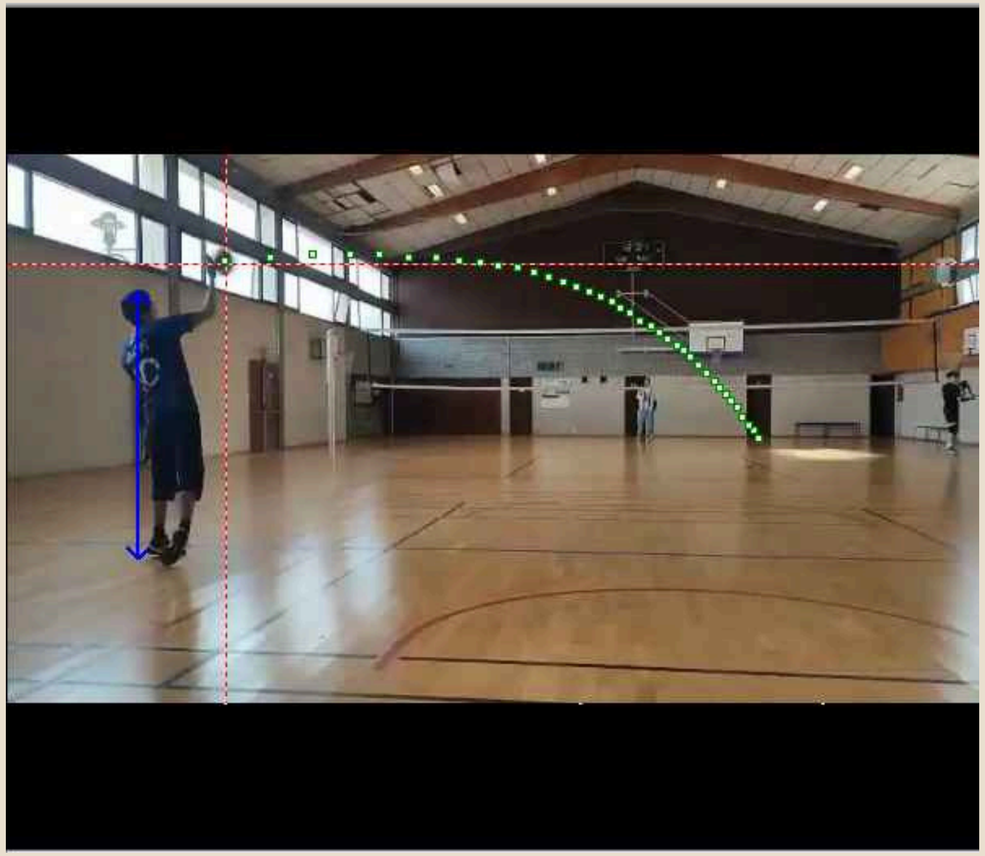


EFFET MAGNUS

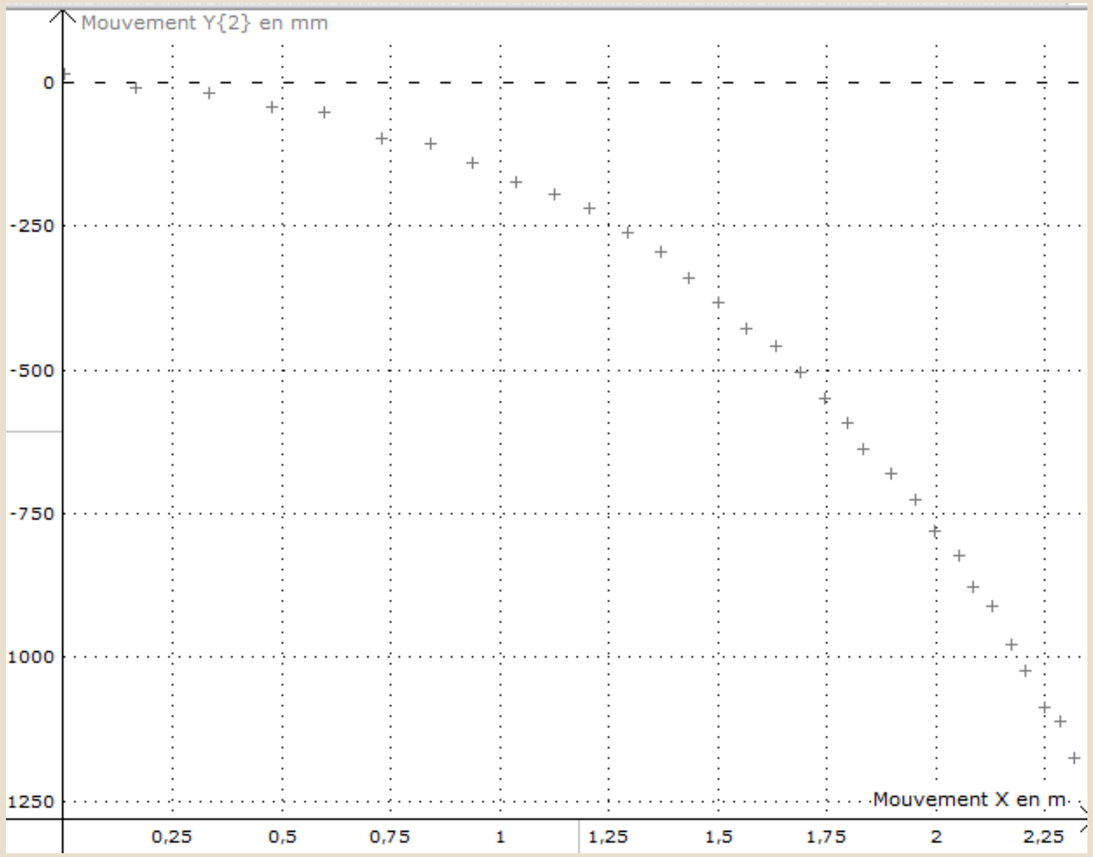




UTILISATION DE LATISPRO
RELEVÉ DES POSITIONS IMAGE PAR IMAGE

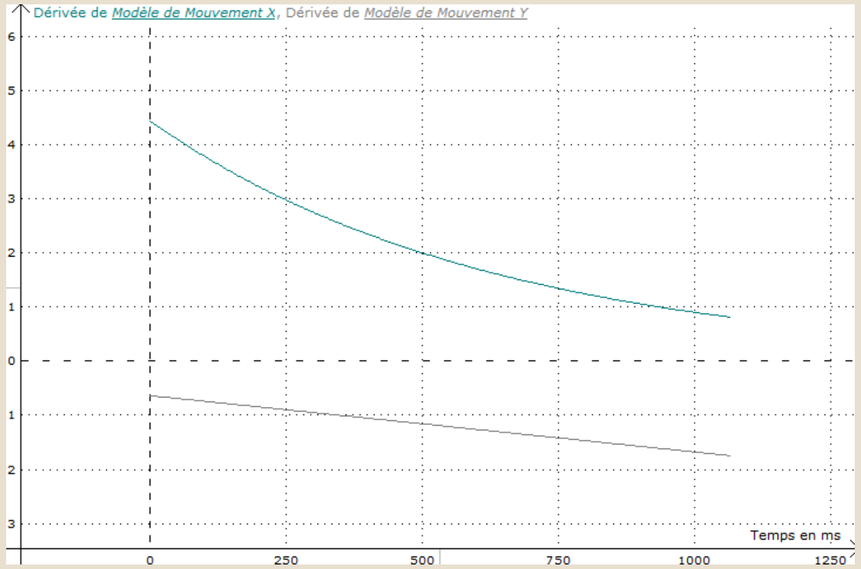


COURBE DU MOUVEMENT Y(X)



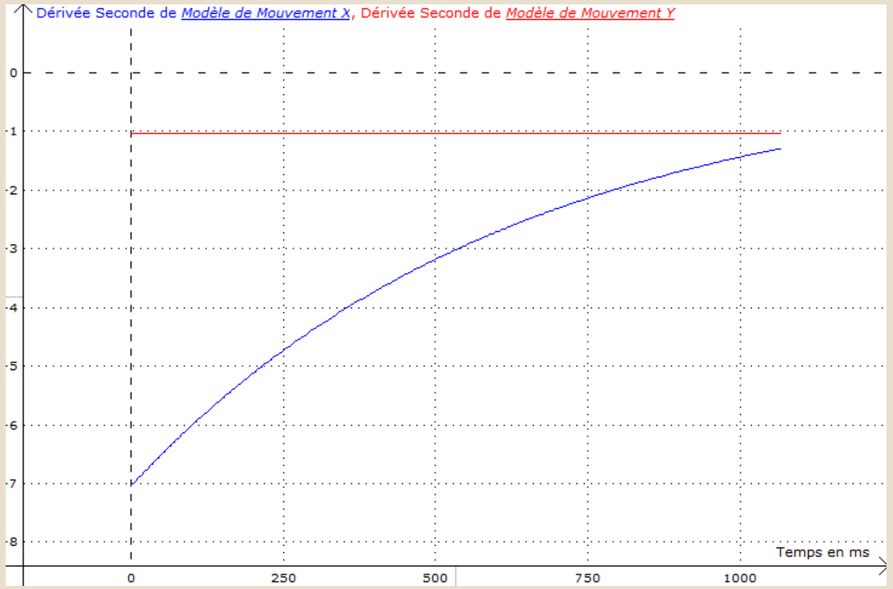
ÉTUDE CINÉMATIQUE

COURBE DE LA DÉRIVÉE DU MOUVEMENT X ET Y



$$v(r) = \sqrt{\left(\frac{dx}{dr}\right)^2 + (dy/dr)^2}$$

COURBE DE LA DÉRIVÉE SECONDE DU MOUVEMENT X ET Y



$$a(r) = dv/dr$$

<insert>
<some_code>
<here>

SIMULATION DE PYTHON

```python
# Constantes
rho = 1.225 # kg/m³
r = 0.105 / 2 # m
A = np.pi * r**2 # m²
m = 0.26 # kg
Cd = 0.5
Cl = 0.2 # à ajuster
g = 9.81

# Conditions initiales
v0 = 20 # m/s
theta = np.radians(10) # angle d'inclinaison vers le haut
w = 90 # rad/s (rotation horizontale autour de l'axe vertical)
dt = 0.01
t_max = 2.5
```

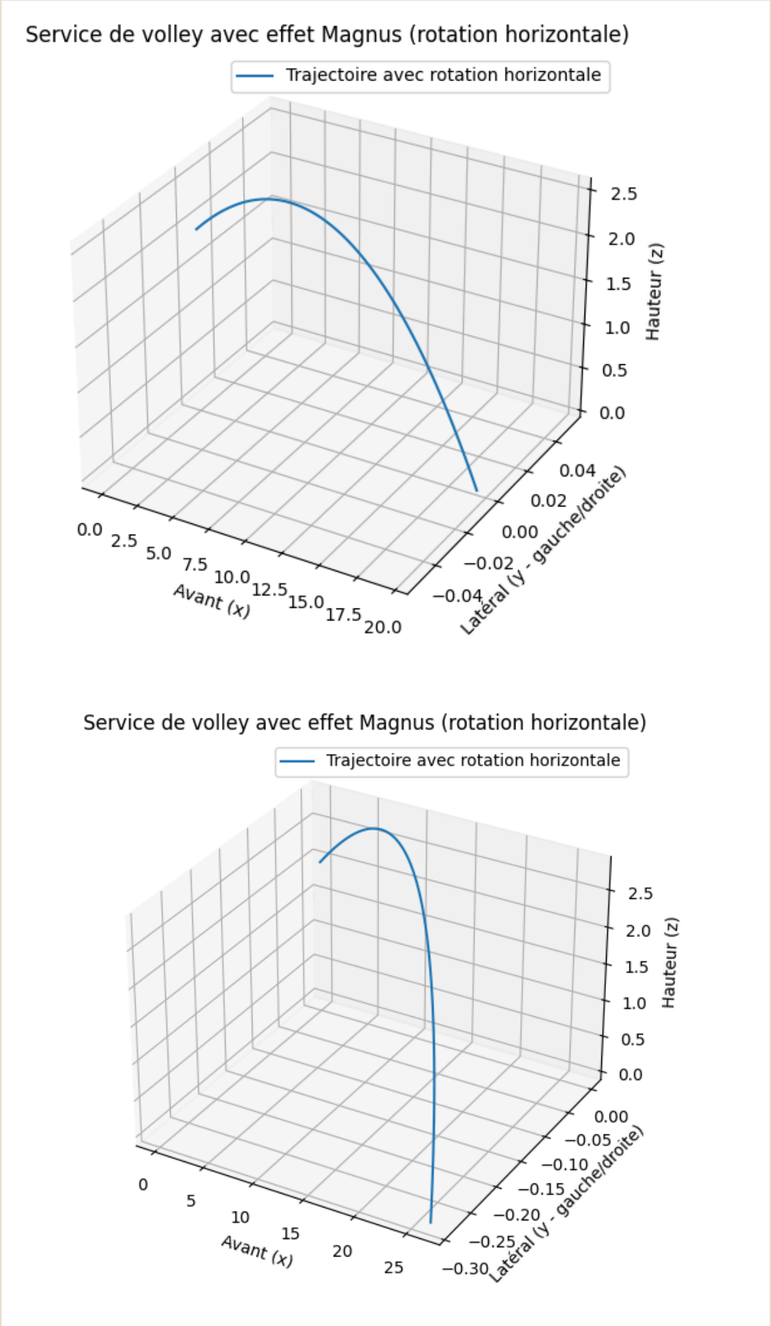
```
>>> (executing cell "" (line 1 of "magnus 2.py"))
Données de la trajectoire (tous les 0.1s environ) :
 Temps (s) |    x (m) |    y (m) |    z (m)
----------------------------------------
      0.00 |    0.000 |    0.000 |    2.000
      0.10 |    1.950 |   -0.002 |    2.290
      0.20 |    3.865 |   -0.008 |    2.478
      0.30 |    5.747 |   -0.017 |    2.565
      0.40 |    7.596 |   -0.029 |    2.553
      0.50 |    9.414 |   -0.044 |    2.444
      0.60 |   11.202 |   -0.063 |    2.240
      0.70 |   12.961 |   -0.084 |    1.941
      0.80 |   14.691 |   -0.109 |    1.550
      0.90 |   16.393 |   -0.136 |    1.068
      1.00 |   18.068 |   -0.166 |    0.496
```

```
>>> (executing cell "" (line 1 of "magnus 2.py"))
Données de la trajectoire (tous les 0.1s environ) :
 Temps (s) |    x (m) |    y (m) |    z (m)
----------------------------------------
      0.00 |    0.000 |    0.000 |    2.000
      0.10 |    1.950 |    0.000 |    2.290
      0.20 |    3.865 |    0.000 |    2.478
      0.30 |    5.747 |    0.000 |    2.565
      0.40 |    7.596 |    0.000 |    2.553
      0.50 |    9.414 |    0.000 |    2.444
      0.60 |   11.202 |    0.000 |    2.240
      0.70 |   12.961 |    0.000 |    1.941
      0.80 |   14.691 |    0.000 |    1.550
      0.90 |   16.393 |    0.000 |    1.068
      1.00 |   18.068 |    0.000 |    0.496
```

```python
# Décomposition vitesse initiale (x: avant, y: latéral, z: vertical)
vx = v0 * np.cos(theta)
vy = 0 # pas de déplacement latéral au début
vz = v0 * np.sin(theta)
x, y, z = 0, 0, 2 # départ au sol à 2m de hauteur

# Stockage
X, Y, Z = [x], [y], [z]

t = 0
while z >= 0 and t <= t_max:
    v = np.sqrt(vx**2 + vy**2 + vz**2)

    # Force de traînée
    Fd = 0.5 * rho * Cd * A * v**2

    # Force Magnus (latérale, vers y)
    Fm = 0.5 * rho * Cl * A * v * w * r

    # Accélérations
    ax = -Fd * (vx / v) / m
    ay = -Fd * (vy / v) / m - Fm / m # Magnus vers la gauche
    az = -g - Fd * (vz / v) / m

    # MàJ vitesses
    vx += ax * dt
    vy += ay * dt
    vz += az * dt

    # MàJ positions
    x += vx * dt
    y += vy * dt
    z += vz * dt

    X.append(x)
    Y.append(y)
    Z.append(z)
    t += dt
```

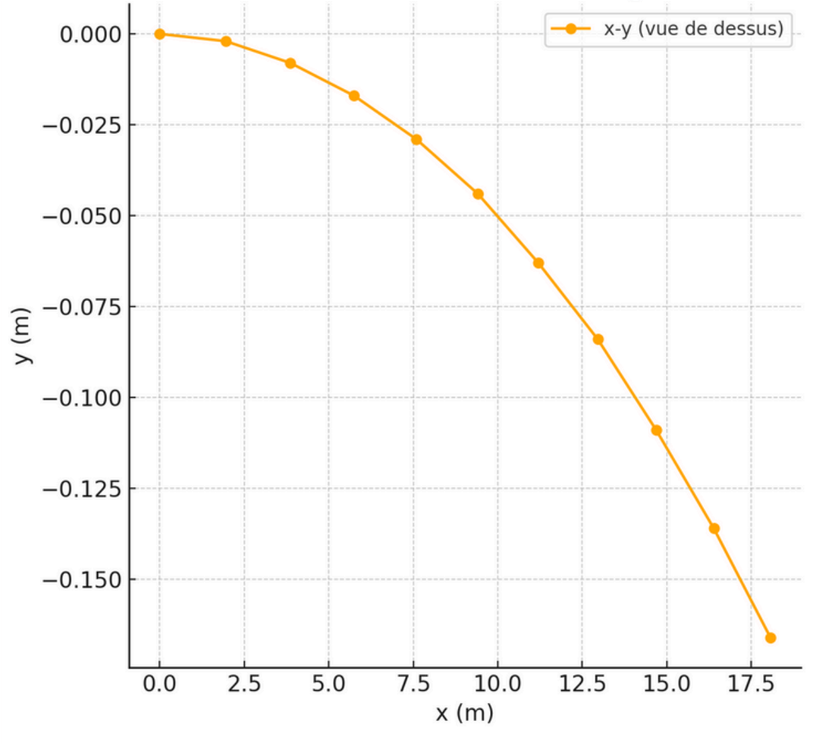
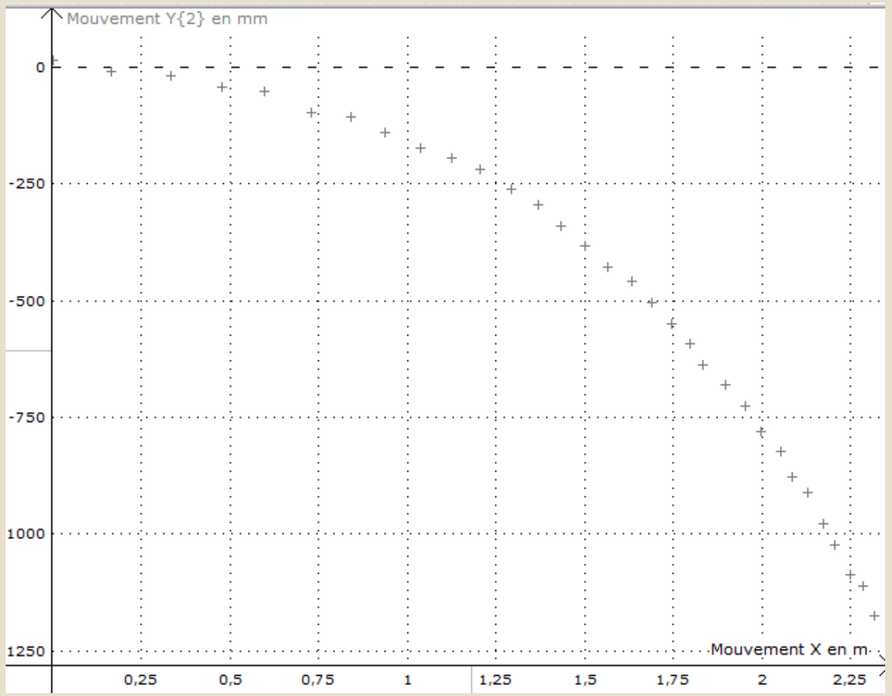
$$V_x(t + \Delta t) = V_x(t) + a_x \cdot \Delta t$$

```python
# Affichage 3D
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(10,6))
ax = fig.add_subplot(111, projection='3d')
ax.plot(X, Y, Z, label='Trajectoire avec rotation horizontale')
ax.set_xlabel('Avant (x)')
ax.set_ylabel('Latéral (y - gauche/droite)')
ax.set_zlabel('Hauteur (z)')
ax.set_title("Service de volley avec effet Magnus (rotation horizontale)")
ax.legend()
plt.show()
# Affichage des points de trajectoire (chaque 0.1s)
print("Données de la trajectoire (tous les 0.1s environ) :")
print(f"{'Temps (s)':>10} | {'x (m)':>8} | {'y (m)':>8} | {'z (m)':>8}")
print("-" * 40)

for i in range(0, len(X), int(0.1 / dt)):
    t_i = i * dt
    print(f"{t_i:10.2f} | {X[i]:8.3f} | {Y[i]:8.3f} | {Z[i]:8.3f}")
```



COMPARAISON DES COURBES

COURBE DU MOUVEMENT Y(X)



COURBE DU MOUVEMENT Y(X)